THE NUMBER OF COLLINEAR TRIPLES IN A $3 \times n$ GRID: LAYMAN'S FLOOR FORMULA (OEIS A057566), AS PRINTED UNTIL OCTOBER 2026, FAILS AT $n = 24$

MATH0

Abstract. Let $a(n)$ be the number of 3-element collinear subsets of the grid $\{0,1,2\} \times \{0,\dots,n-1\}$ (OEIS A057566). The OEIS entry, submitted by John W. Layman in October 2000, records the conjecture $a(n) = 5\lfloor(2n^3-3n^2-n)/24\rfloor+\lfloor(2(n-1)^3-3(n-1)^2-(n-1))/24\rfloor+n$. We show that this identity is false for $n = 24$: the 3×24 grid has 6360 collinear triples, while the formula gives 6365. The falsity is not new in substance, since $a(24) = 6360$ is listed in the entry's own data and a proved closed form for $a(n)$ follows from an argument recorded in OEIS A262402; we know of no explicit published refutation, and this is the first formal proof of the known result. The refutation is verified in Lean 4 against the frozen, fidelity-reviewed statement that math0 formalized from the OEIS entry, whose definition of a is checked by the kernel against the first 25 listed terms $a(0),\dots,a(24)$.

1. Introduction

For $n \geq 0$ let $G_n = \{0,1,2\} \times \{0,1,\dots,n-1\} \subset \mathbb{Z}^2$ be the $3 \times n$ grid of lattice points, and let $a(n)$ be the number of 3-element subsets of G_n whose points lie on a common line. This is the sequence A057566 of the OEIS [5], "Number of collinear triples in a 3 X n rectangular grid", with offset 0 and first terms $0, 1, 2, 8, 20, 43, 78, 130, 200, 293, 410, \dots$ The entry was submitted by John W. Layman on October 4, 2000. Its FORMULA section contains the conjecture

$$a(n) \stackrel{?}{=} 5f(n) + f(n-1) + n, \qquad f(m) = \Big\lfloor \frac{2m^3 - 3m^2 - m}{24} \Big\rfloor, \tag{1}$$

with the remark that it "fits all of the listed terms". The line was still labelled "Conjecture" in revision #27 of the entry (February 22, 2024), the text formalized here; it has since been corrected (see the note below). We prove that (1) fails at $n = 24$ (Theorem 1).

Note added on October 26, 2026. On October 21, 2026, after the statement studied here was frozen, the formula in the entry was corrected by Alper Ferudun to

$$a(n) = 5\Big\lfloor \frac{2n^3 - 3n^2 - 2n + 3}{24} \Big\rfloor + \Big\lfloor \frac{2(n-1)^3 - 3(n-1)^2 - 2(n-1) + 3}{24} \Big\rfloor + n.$$

Our theorem concerns only the formula as printed before that correction, quoted in Section 2. The corrected formula is not refuted: it agrees with a brute-force count of collinear triples for $n \leq 60$ and with the closed form of Proposition 5 for $n \leq 20000$. A proof of the corrected formula is not part of this paper.

Prior work and status. The refutation is elementary and its ingredients are already in the OEIS; we credit them precisely.

- The data section of A057566 lists $a(0),\dots,a(42)$, including $a(24) = 6360$, and the entry links a table of $a(n)$ for $n \leq 10000$ by Paolo Xausa [5]. Formula (1) gives 6365 at $n = 24$, so the remark "fits all of the listed terms" is contradicted by the entry's own data. We

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do not know how many terms were listed in 2000; the formula agrees with $a(n)$ for $3 \le n \le 23$.

- On May 23, 2010, R. J. Mathar added to A057566 the closed form $a(n) = n^3/2 - n^2 + n + (1-(-1)^n)/4$, the linear recurrence with signature $(3, -2, -2, 3, -1)$ and the generating function $x(1-x+4x^2+2x^3)/((1+x)(x-1)^4)$ [5]. The entry states these without proof.
- In OEIS A262402 [6] (triangles with vertices in a $3 \times n$ grid), N. J. A. Sloane gave on December 16, 2017 a proof of a formula of Jared Nash; see also Nash's note linked there [4]. The proof shows that the number of collinear triples with one point in each row of the $3 \times n$ grid is $n + 2\lfloor (n-1)^2/4 \rfloor$. Together with the $3\binom{n}{3}$ triples inside a single row (a triple with exactly two points in one row is never collinear), this gives a proof of $a(n) = 3\binom{n}{3} + n + 2\lfloor (n-1)^2/4 \rfloor$, which equals Mathar's closed form and gives $a(24) = 6360$.
- The triangle A334704 [7] (N. J. A. Sloane, 2020, with rows 6 onwards from Tom Duff, who computed the values for grids up to 50×50) counts collinear triples in $n \times k$ grids ($1 \le k \le n$); in the full symmetric array of which it is the lower half, row 3 (the $3 \times k$ grids, $k \ge 1$) is A057566, as the entry notes.

The math0 referees found no explicit statement, in the OEIS or elsewhere (arXiv and general web searches), that Layman's conjecture is false. Comparing (1) with the listed term $a(24)$, or with Mathar's closed form, is a one-line computation, so we classify the result as the *first formal proof of a known result*, not as a new result.

What is proved formally. The Lean proof shows that the identity (1) does not hold for all $n \ge 3$, by exhibiting $n = 24$. It does not formalize the closed form for $a(n)$ (Proposition 5) or the fact that (1) fails for every $n \ge 24$ (Remark 7); both are proved here on paper only.

2. Statement

The FORMULA section of A057566 [5] contains, verbatim, the unsigned line

```
Conjecture:  a(n) = 5*floor((2n^3 - 3n^2 - n)/24) + floor((2(n-1)^3 -
3(n-1)^2 - (n-1))/24) + n, which fits all of the listed terms.
```

It is the first FORMULA line of the entry, whose author line reads `_John W. Layman_, Oct 04 2000`; we attribute the conjecture to him, as the math0 statement file does.

Theorem 1. *Let $a(n)$ be the number of collinear 3-element subsets of $\{0,1,2\} \times \{0,\dots,n-1\}$, and let $f(m) = \lfloor (2m^3 - 3m^2 - m)/24 \rfloor$ for $m \in \mathbb{Z}$. Then*

$$a(24) = 6360 \neq 6365 = 5f(24) + f(23) + 24.$$

In particular, it is false that $a(n) = 5f(n) + f(n-1) + n$ for all $n \ge 3$.

The range of n. The conjecture names no range for n, and the sequence has offset 0. For $n \le 2$ some of the arguments of the floors are negative: $(2m^3 - 3m^2 - m)/24$ equals $-4/24$ at $m = -1$ and $-2/24$ at $m = 1$. With true floors, (1) gives $-1, -4, 1$ at $n = 0, 1, 2$ instead of $0, 1, 2$; with rounding toward zero it gives exactly $0, 1, 2$, and then agrees with $a(0), \dots, a(23)$. A refutation at $n \le 2$ would thus rest on a rounding convention at degenerate grid sizes. For $n \ge 3$ both arguments are nonnegative ($2m^3 - 3m^2 - m = m\big(m(2m-3) - 1\big) > 0$ for $m \ge 2$), so every reading of "floor" gives the same claim. The math0 statement therefore asserts (1) for $n \ge 3$ only. This is the weakest faithful reading, and refuting it refutes every reading; the literal reading for all $n \ge 0$ is recorded separately as `statement_all`, with a proof that it implies `statement`.

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A057566.lean`, reviewed for fidelity, and frozen (SHA-256 `218f1d78...6ec5`) before any proof attempt. The definitions and the statement read as follows (verbatim; the copyright header, the import, docstrings, blank lines, the fast evaluator of Section 3 with its auxiliary multiset predicate `Q`, all theorems, and the variant `statement_all`, which is `statement` without the hypothesis `3 ≤ n`, are omitted):

```
namespace Math0.Statements.A057566
def grid (n : ℕ) : Finset (ℕ × ℕ) := Finset.range 3 ×ˢ Finset.range n
def cross (p q r : ℕ × ℕ) : ℤ :=
  ((q.1 : ℤ) - p.1) * ((r.2 : ℤ) - p.2) - ((q.2 : ℤ) - p.2) * ((r.1 : ℤ) - p.1)
def IsCollinear (s : Finset (ℕ × ℕ)) : Prop := ∀ p ∈ s, ∀ q ∈ s, ∀ r ∈ s, cross p q r = 0
instance : DecidablePred IsCollinear := fun s => by unfold IsCollinear; infer_instance
def a (n : ℕ) : ℕ := ((grid n).powersetCard 3 |>.filter IsCollinear).card
def statement : Prop :=
  ∀ n : ℕ, 3 ≤ n → (a n : ℤ) =
    5 * ⌊(2 * (n : ℚ) ^ 3 - 3 * (n : ℚ) ^ 2 - (n : ℚ)) / 24⌋
      + ⌊(2 * ((n : ℚ) - 1) ^ 3 - 3 * ((n : ℚ) - 1) ^ 2 - ((n : ℚ) - 1)) / 24⌋ + n
end Math0.Statements.A057566
```

A point is a pair (row, column) of natural numbers, and `grid n` is G_n. For points p, q, r the integer

$$\mathrm{cr}(p,q,r) = (q_1 - p_1)(r_2 - p_2) - (q_2 - p_2)(r_1 - p_1)$$

is twice the signed area of the triangle pqr, so it vanishes exactly when p, q, r lie on a common line. A finite set is `IsCollinear` if cr vanishes on every three of its points (repetitions allowed), and `a n` is the number of 3-element subsets of `grid n` with this property, i.e. the number of collinear triples. The floors are `Int.floor` of rational numbers, and $n - 1$ is computed in $\mathbb{Q}$, so it is not truncated at 0. Among other lemmas, the statement file proves (statements shown, proofs omitted):

```
theorem isCollinear_triple (p q r : ℕ × ℕ) :
    IsCollinear {p, q, r} ↔ cross p q r = 0
theorem a_eq_fastA (n : ℕ) : a n = fastA n
theorem a_data : (List.range 25).map (fun i => a (i + 0)) =
    [0, 1, 2, 8, 20, 43, 78, 130, 200, 293, 410, 556, 732, 943, 1190, 1478, 1808, 2185,
      2610, 3088, 3620, 4211, 4862, 5578, 6360]
theorem statement_all_imp : statement_all → statement
theorem statement_iff_ediv : statement ↔ ∀ n : ℕ, 3 ≤ n → (a n : ℤ) =
    5 * ((2 * (n : ℤ) ^ 3 - 3 * (n : ℤ) ^ 2 - (n : ℤ)) / 24)
      + ((2 * ((n : ℤ) - 1) ^ 3 - 3 * ((n : ℤ) - 1) ^ 2 - ((n : ℤ) - 1)) / 24) + n
```

The lemma `isCollinear_triple` shows that for a 3-element set the collinearity condition is the single equation $\mathrm{cr}(p,q,r) = 0$. The data check `a_data`, proved by `decide +kernel` through `a_eq_fastA`, confirms that the definition reproduces the first 25 terms of the OEIS data, $a(0), \ldots, a(24)$; this anchors the definition to the entry independently of the conjecture.

3. Proof

The proof follows the Lean file `lean/Math0/Frontier/OEIS/A057566.lean`. It has two steps: the left-hand side $a(24)$ is computed by the kernel through a verified fast counter, and the right-hand side is evaluated by integer arithmetic.

3.1. Counting the triples: a verified evaluator.

The definition of `a` enumerates the 3-element subsets of a finset, which is too slow for evaluation in the kernel. The statement file therefore also defines a list-based counter (verbatim, docstrings omitted):

```
def collB (p q r : ℕ × ℕ) : Bool :=
  Nat.beq (q.1 * r.2 + q.2 * p.1 + p.2 * r.1) (q.2 * r.1 + q.1 * p.2 + p.1 * r.2)
def cnt2 (p q : ℕ × ℕ) : List (ℕ × ℕ) → ℕ
  | [] => 0
  | r :: l => cnt2 p q l + (if collB p q r then 1 else 0)
def cnt1 (p : ℕ × ℕ) : List (ℕ × ℕ) → ℕ
  | [] => 0
  | q :: l => cnt1 p l + cnt2 p q l
def cnt : List (ℕ × ℕ) → ℕ
  | [] => 0
  | p :: l => cnt l + cnt1 p l
def pts (n : ℕ) : List (ℕ × ℕ) := List.range 3 ×ˢ List.range n
def fastA (n : ℕ) : ℕ := cnt (pts n)
```

Thus $\mathrm{cnt}(\ell)$ is the number of triples of positions $i < j < k$ in the list ℓ whose entries p, q, r satisfy `collB`, and `pts n` lists the $3n$ points of G_n, each once.

Lemma 2 (`collB_iff`, `isCollinear_triple`, `Q_triple`)**.** *For points $p, q, r \in \mathbb{N}^2$,* `collB`(p, q, r) *is true if and only if* $\mathrm{cr}(p, q, r) = 0$*; and the finset $\{p, q, r\}$ (respectively the multiset with elements p, q, r) satisfies* `IsCollinear` *(respectively its multiset analogue* `Q`*, with the same definition) if and only if* $\mathrm{cr}(p, q, r) = 0$*.*

Proof. Expanding, $\mathrm{cr}(p, q, r) = (q_1 r_2 + q_2 p_1 + p_2 r_1) - (q_2 r_1 + q_1 p_2 + p_1 r_2)$, and both brackets are natural numbers; `collB` tests their equality. For the second claim, one direction is the instance (p, q, r) of the definition. Conversely, cr vanishes whenever two of its arguments are equal and changes at most its sign under a permutation of its arguments, so $\mathrm{cr}(p, q, r) = 0$ implies $\mathrm{cr}(x, y, z) = 0$ for all $x, y, z \in \{p, q, r\}$. The argument is the same for the finset and the multiset; in Lean the 27 cases are closed by `ring1` or `linear_combination` $\pm h$. □

Lemma 3 (`a_eq_fastA`)**.** *For every $n \geq 0$, $a(n) =$* `fastA`(n)*.*

Proof. The finset `grid n` has underlying multiset equal to the list `pts n` (`grid_val`, by `rfl`), and `a n` counts the elements of its powerset of size 3 that are collinear. Under the coercion from lists to multisets, the 3-element sub-multisets of a list ℓ are the images of the sublists of ℓ of length 3, `List.sublistsLen 3` ℓ (`Multiset.powersetCard_coe`, together with `Finset.map_val_val_powersetCard` to pass from finsets to multisets). So $a(n)$ is the number of length-3 sublists of `pts n` that are collinear. The recursion

$$\texttt{sublistsLen}\,(k+1)\,(p :: \ell) = \texttt{sublistsLen}\,(k+1)\,\ell \;{+\!\!+}\; \texttt{map}\,(p :: \cdot)\,(\texttt{sublistsLen}\,k\,\ell)$$

(`List.sublistsLen_succ_cons`) splits the sublists by whether they use the head p. By induction on ℓ this identifies the number of collinear sublists of length 3, of length 2 preceded by p, and of length 1 preceded by p, q with $\mathrm{cnt}(\ell)$, $\mathrm{cnt1}(p, \ell)$ and $\mathrm{cnt2}(p, q, \ell)$ respectively (`countP_three`, `countP_two`, `countP_one`). Collinearity of a sub-multiset is tested by `Q`, and at the innermost level, in `countP_one`, the test on the multiset $[p, q, r]$ is $\mathrm{cr}(p, q, r) = 0$ by `Q_triple`, i.e. `collB` by Lemma 2. □

Lemma 4 (`a_24`)**.** $a(24) = 6360$*.*

Proof. By Lemma 3, $a(24) =$ `fastA`(24). The equation `fastA`$(24) = 6360$ between natural numbers is decidable, and `decide +kernel` asks the Lean kernel to reduce the decision procedure `Nat.decEq (fastA 24) 6360` to `isTrue`. The kernel does so by evaluating the nested recursion over the $\binom{72}{3} = 59640$ triples of points of G_{24}, with built-in arithmetic on natural-number literals (including `Nat.beq`). No compiled code is trusted and no axiom is added; the certificate is the evaluation itself. □

For the reader we also give a short proof of the value 6360, and of the closed form, which the Lean proof does not use.

Proposition 5 (not formalized)**.** *For every $n \geq 0$,*

$$a(n) = 3\binom{n}{3} + \left\lceil \frac{n}{2} \right\rceil^2 + \left\lfloor \frac{n}{2} \right\rfloor^2 = \frac{n^3}{2} - n^2 + n + \frac{1-(-1)^n}{4}.$$

Proof. Classify a 3-element subset of G_n by the rows of its points. If all three points lie in one row, they are collinear; this gives $3\binom{n}{3}$ triples. If two points $p \neq q$ lie in row i and the third point r in row $i' \neq i$, then $\mathrm{cr}(p,q,r) = -(q_2 - p_2)(i' - i) \neq 0$, so the triple is not collinear. If the points are $(0,x)$, $(1,y)$, $(2,z)$, then $\mathrm{cr} = (z-x) - 2(y-x) = x + z - 2y$, so the triple is collinear if and only if $y = (x+z)/2$. Such a y exists exactly when $x \equiv z \pmod 2$, and then $0 \leq y \leq n-1$ automatically. Hence these triples correspond bijectively to the pairs $(x,z) \in \{0,\dots,n-1\}^2$ of equal parity, of which there are $\lceil n/2 \rceil^2 + \lfloor n/2 \rfloor^2 = (n^2 + [n \text{ odd}])/2$. Adding $3\binom{n}{3} = (n^3 - 3n^2 + 2n)/2$ gives the second expression. □

At $n = 24$ this gives $a(24) = 3 \cdot 2024 + 144 + 144 = 6360$, in agreement with Lemma 4, with the OEIS data, and with the formula $3\binom{n}{3} + n + 2\lfloor (n-1)^2/4 \rfloor$ that follows from Sloane's argument in A262402 [6].

3.2. **The right-hand side.**

Lemma 6 (`statement_iff_ediv`)**.** *For every integer N, $\lfloor N/24 \rfloor$ (the floor of the rational number $N/24$) equals the integer quotient $N \,/\, 24$ in Lean. Consequently* `statement` *is equivalent to the same identity with the two floors replaced by integer division by* 24*.*

Proof. For a positive divisor, Lean's integer division `/` on $\mathbb{Z}$ rounds toward $-\infty$, so it agrees with the floor of the rational quotient (`Rat.floor_intCast_div_natCast`). Apply this to $N = 2m^3 - 3m^2 - m$ with $m = n$ and $m = n-1$, after moving the casts from $\mathbb{Q}$ to $\mathbb{Z}$. □

Proof of Theorem 1. Suppose the identity held for all $n \geq 3$. Write $N_m = 2m^3 - 3m^2 - m$ and $x \operatorname{div} 24$ for the integer quotient. By Lemma 6, at $n = 24$ the identity reads

$$a(24) = 5\,(N_{24} \operatorname{div} 24) + (N_{23} \operatorname{div} 24) + 24.$$

Here $N_{24} = 2 \cdot 24^3 - 3 \cdot 24^2 - 24 = 27648 - 1728 - 24 = 25896 = 24 \cdot 1079$, so $f(24) = 1079$, and $N_{23} = 2 \cdot 23^3 - 3 \cdot 23^2 - 23 = 24334 - 1587 - 23 = 22724 = 24 \cdot 946 + 20$, so $f(23) = 946$. The right-hand side is $5 \cdot 1079 + 946 + 24 = 6365$, whereas $a(24) = 6360$ by Lemma 4. This contradiction proves the theorem. In Lean, after rewriting with `statement_iff_ediv` and `a_24`, the hypothesis at $n = 24$ is an equation between closed integer expressions, $6360 = 5 \cdot (N_{24} \,/\, 24) + (N_{23} \,/\, 24) + 24$ with N_{24} and N_{23} written out as above, which `norm_num` evaluates to $6360 = 6365$ and hence to `False`. □

Remark 7 (failure for all $n \geq 24$; not formalized)**.** Write $L(n) = 5f(n) + f(n-1) + n$ and $\{t\} = t - \lfloor t \rfloor$. Expanding,

$$\frac{5(2n^3 - 3n^2 - n) + \big(2(n-1)^3 - 3(n-1)^2 - (n-1)\big)}{24} + n = \frac{n^3}{2} - n^2 + \frac{5n}{4} - \frac{1}{6},$$

so by Proposition 5

$$L(n) - a(n) = \frac{n}{4} - \frac{1}{6} - \frac{1-(-1)^n}{4} - 5\theta_n - \theta_{n-1}, \qquad \theta_m = \left\{ \frac{2m^3 - 3m^2 - m}{24} \right\} \in [0,1).$$

Hence $n/4 - 20/3 < L(n) - a(n) \leq n/4 - 1/6$, so $L(n) > a(n)$ for all $n \geq 27$, and $L(n) - a(n) = n/4 + O(1)$. Together with the values $L(n) - a(n) = 5, 1, 5$ at $n = 24, 25, 26$, this shows that (1) fails for every $n \geq 24$. It holds for $3 \leq n \leq 23$ (a finite check, which we did by computer), so

$n = 24$ is the least counterexample with $n \geq 3$. The failure is therefore not a rounding artefact: the cubic and quadratic terms of (1) are correct, but its linear term $5n/4$ should be n.

4. Machine verification

Files and declarations. The proof is in `lean/Math0/Frontier/OEIS/A057566.lean` in the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file `lean/Math0/Statements/OEIS/A057566.lean` (Section 2) and the audit commands in `lean/Math0/Audit.lean`. The main declaration is `Math0.Frontier.A057566.result`, of type `¬ Math0.Statements.A057566.statement`; the only auxiliary declaration is the private lemma `a_24`. The proof reads, verbatim (header, imports, module docstring and docstrings omitted):

```
namespace Math0.Frontier.A057566
open Math0.Statements.A057566
private theorem a_24 : a 24 = 6360 := by
  rw [a_eq_fastA]
  decide +kernel
theorem result : ¬ statement := by
  rw [statement_iff_ediv]
  intro h
  have h24 := h 24 (by norm_num)
  rw [a_24] at h24
  norm_num at h24
end Math0.Frontier.A057566
#math0_settles Math0.Frontier.A057566.result Math0.Statements.A057566.statement
```

Neither file uses `sorry`, `native_decide`, custom axioms or `set_option`.

Audit. The command `#math0_settles`, defined in `lean/Math0/Audit.lean` for statements that math0 formalizes itself, fails unless (1) `statement` is a `Prop`-valued definition without universe parameters; (2) `statement` and `result` depend on no axioms besides `propext`, `Classical.choice` and `Quot.sound` (this rejects `sorryAx`, `Lean.ofReduceBool` from `native_decide`, and custom axioms); and (3) the type of `result` is syntactically `statement` or `¬ statement`, up to metadata annotations, with no unfolding. Here it reports

```
math0 audit: 'Math0.Statements.A057566.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A057566.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A057566.result' refutes 'Math0.Statements.A057566.statement'
```

Toolchain and reproduction. We use Lean 4 [3] `v4.33.1` and Mathlib [2] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [1] at commit `c77fdd4`, which this result does not use, since the statement is a math0 formalization of an OEIS conjecture and not part of formal-conjectures. To reproduce, run `scripts/verify.sh` from the repository root: it fetches the Mathlib build cache, builds all math0 results with `lake build` (re-running every audit command), checks that the pinned `formal_conjectures` and `mathlib` checkouts are unmodified, and reports the number of passed statement checks. The single file can be checked with `lake env lean Math0/Frontier/OEIS/A057566.lean` in the directory `lean`; on our machine this took about one minute of CPU time (user and system), including loading Mathlib and the kernel evaluation of Lemma 4.

References

[1] M. Firsching, P. Lezeau, S. Mercuri, M. Z. Horváth, Y. Dillies, C. Sönne, E. Wieser, F. Zhang, T. Hubert, B. Agüera y Arcas, and P. Kohli, *Formal Conjectures: An Open and Evolving Benchmark for Verified Discovery in Mathematics*, arXiv:2605.13171 [cs.AI], 2026. Repository: `https://github.com/google-deepmind/formal-conjectures`.

[2] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.

[3] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635.

[4] J. Nash, *Formula for number of triangles in a 3 X n grid*, note linked from OEIS A262402, `https://oeis.org/A262402/a262402.pdf`.

[5] The OEIS Foundation Inc., *Number of collinear triples in a $3 \times n$ rectangular grid*, entry A057566 (J. W. Layman, 2000) in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A057566`; revision #27, February 22, 2024.

[6] The OEIS Foundation Inc., *Number of triangles that can be formed from the points of a $3 \times n$ grid*, entry A262402 (R. Pan, 2015; revised by N. J. A. Sloane, 2017) in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A262402`.

[7] The OEIS Foundation Inc., *Triangle read by rows: number of ways to choose three collinear points from an $n \times k$ grid of points*, entry A334704 (N. J. A. Sloane and T. Duff, 2020) in The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A334704`.

Produced by Claude (an AI model developed by Anthropic) in the math0 project